SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Sowmiya. D (192511063) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sowmiya. D

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

    i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

    v. All doctors must be assigned exactly one shift

    Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule
2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

    Constraints:

    i. Movement allowed: Up, Down, Left, Right ii. Cost of each move = 1
    iii. Cannot pass through obstacles
    iv. Use Manhattan Distance heuristic

    Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts

Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. **Constraint Satisfaction Problem (CSP)** using **Backtracking Search**.

Variables

- Doctors: D1, D2, D3

Domain (Possible Shifts)

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. **D1 ≠ Night**
2. **D2 works before D3** (Morning < Afternoon < Night)
3. **Only one doctor per shift**
4. **D3 ≠ Morning**
5. **Each doctor gets exactly one shift**

Backtracking Search

Start with an empty assignment.

Initial Assignment: {}

Step 1: Assign D1

Possible values:

- Morning ✓
- Afternoon ✓
- Night ✗ (Constraint i violated)

Choose the first valid value: D1 = Morning

Current Assignment: {D1 = Morning}

Constraint Check:

- D1 not Night ✓

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Try D2 = Afternoon

Current Assignment: {D1 = Morning, D2 = Afternoon}

Constraint Check:

- Unique shifts ✓
- D2 before D3 (not checked fully until D3 assigned)

Proceed.

Step 3: Assign D3

Remaining shift:

- Night (Try)

D3 = Night

Current Assignment: {D1 = Morning, D2 = Afternoon, D3 = Night}

Constraint Check:

- D3 not Morning ✓
- One doctor per shift ✓
- D2 before D3
    - Afternoon < Night ✓
- Every doctor assigned exactly one shift ✓

All constraints satisfied.

Hence, a solution is found.

Search Tree

```
        {}
        |
   D1 = Morning
        |
   D2 = Afternoon
        |
```

D3 = Night

|

Solution Found

Final Valid Schedule

- **Morning → D1**
- **Afternoon → D2**
- **Night → D3**
-

2. Robot Navigation Using BFS Search Algorithm (10 Marks)

Given

A robot moves in a 5 × 5 grid.

Constraints

1. Movement allowed: Up, Down, Left, Right
2. Cost of each move = 1
3. Robot cannot move through obstacles.
4. Manhattan Distance heuristic is given (Note: BFS does not use heuristics; Manhattan Distance is used in Greedy Best-First Search or A*. Since the question specifically asks for BFS, the heuristic is not used during the search.)

Grid Representation

	C1	**C2**	**C3**	**C4**	**C5**
R1	**S**	**.**	**.**	**X**	**.**
R2	**.**	**X**	**.**	**X**	**.**
R3	**.**	**X**	**.**	**.**	**.**
R4	**.**	**.**	**X**	**X**	**.**
R5	**X**	**.**	**.**	**.**	**G**

Where:

- **S = Start = (1,1)**
- **G = Goal = (5,5)**

- **X = Obstacle**
- **. = Free cell**

BFS Algorithm

Step 1: Initialization

Queue = [(1,1)]

Visited = {(1,1)}

Step-by-Step Node Expansion

Step	Expanded Node	New Nodes Added	Queue after Update
1	(1,1)	(1,2), (2,1)	[(1,2), (2,1)]
2	(1,2)	(1,3)	[(2,1), (1,3)]
3	(2,1)	(3,1)	[(1,3), (3,1)]
4	(1,3)	(2,3)	[(3,1), (2,3)]
5	(3,1)	(4,1)	[(2,3), (4,1)]
6	(2,3)	(3,3)	[(4,1), (3,3)]
7	(4,1)	(4,2)	[(3,3), (4,2)]
8	(3,3)	(3,4)	[(4,2), (3,4)]
9	(4,2)	(5,2)	[(3,4), (5,2)]
10	(3,4)	(3,5)	[(5,2), (3,5)]
11	(5,2)	(5,3)	[(3,5), (5,3)]
12	(3,5)	(4,5), (2,5)	[(5,3), (4,5), (2,5)]
13	(5,3)	(5,4)	[(4,5), (2,5), (5,4)]
14	(4,5)	(5,5) Goal Found	[(2,5), (5,4), (5,5)]

BFS Search Tree

```
(1,1)
├── (1,2)
│   └── (1,3)
│       └── (2,3)
│           └── (3,3)
│               └── (3,4)
│                   └── (3,5)
│                       └── (4,5)
│                           └── (5,5) Goal
└── (2,1)
    └── (3,1)
        └── (4,1)
            └── (4,2)
                └── (5,2)
                    └── (5,3)
                        └── (5,4)
```

Optimal Path Found by BFS: (1,1) → (2,1) → (3,1) → (4,1) → (4,2) → (5,2) → (5,3) → (5,4) → (5,5)

Path Cost

Number of moves = 8

Cost per move = 1

Total Cost = 8 × 1 = 8

Manhattan Distance Heuristic (for Goal at (5,5))

The Manhattan Distance is:

$$\boldsymbol{h(n) = | x - 5 | + | y - 5 |}$$

For the start node:

$$\boldsymbol{h(1,1) = | 1 - 5 | + | 1 - 5 | = 4 + 4 = 8}$$

3. Autonomous Rescue Robot in a Disaster-Affected Building (10 Marks)

a) Analysis of the Constraints and Their Impact on Decision-Making

The rescue robot must operate in a partially observable environment, so it cannot see the entire building at once. Each constraint affects the robot's decisions as follows:

Constraint	**Effect on Decision-Making**
Robot can move Up, Down, Left, Right	The robot considers only four possible moves from each position.
Blocked cells (obstacles)	The robot cannot enter obstacle cells and must find an alternate path.
Limited sensing (adjacent cells only)	The robot has incomplete knowledge and updates its map as it explores.
Normal movement cost = 1	The robot tries to minimize the total travel cost.
Risky zones have additional cost of 2	The robot avoids risky areas unless they provide the lowest overall cost.
Dynamic knowledge update	The robot replans its route whenever it discovers new obstacles or risks.

b) Solution Approach Using Uniform Cost Search (UCS) with Online Search

Why Uniform Cost Search?

- Finds the minimum-cost path.
- Considers different movement costs.
- Suitable when risky zones have higher costs.
- Can be combined with online exploration in a partially observable environment.

Algorithm Steps

Step 1: Start from position S.

Step 2: Observe adjacent cells using sensors.

Step 3: Add all reachable neighboring cells to the priority queue with their cumulative costs.

Step 4: Select the node with the lowest total cost.

Step 5: If a new obstacle or risky zone is discovered:

- Update the map.
- Recalculate the best path.

Step 6: Repeat until the robot reaches the survivor G.

Flow of the Algorithm

Start (S)

↓

Sense Adjacent Cells

↓

Update Environment Map

↓

Expand Lowest-Cost Node (UCS)

↓

Obstacle Found?

/ \

Yes **No**

↓ ↓

Update Map **Continue Search**

↓

Goal Reached (G)

c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an Intelligent Agent because it:

- **Perceives the environment using sensors.**
- **Processes information.**
- **Makes decisions.**
- **Performs actions using actuators.**

PEAS Representation

Component	Description
Performance Measure	Reach survivor safely with minimum cost
Environment	Disaster-affected building
Actuators	Wheels/Motors
Sensors	Cameras, proximity sensors, obstacle detectors

2. Rationality

A rational agent always selects the action that maximizes performance.

The robot behaves rationally by:

- **Avoiding obstacles.**
- **Avoiding risky zones whenever possible.**
- **Choosing the least-cost path.**
- **Updating decisions when new information becomes available.**

3. Environment Type

Property	Description
Partially Observable	Robot sees only adjacent rooms.
Dynamic	Environment may change due to falling debris or blocked paths.
Sequential	Current action affects future decisions.
Discrete	Building consists of grid cells.
Single Agent	Only one rescue robot performs the task.
Unknown	Robot learns the map while exploring.

d) Justification for the Chosen Approach

The combination of Uniform Cost Search (UCS) and Online Search is the most suitable because:

- **UCS guarantees the lowest-cost path.**
- **It considers different movement costs, including risky zones.**

- **Online search allows the robot to operate without complete knowledge of the environment.**
- **The robot can dynamically update its map when new obstacles are detected.**
- **The approach ensures both safe navigation and minimum rescue cost.**

Compared to other algorithms:

- **BFS ignores varying movement costs.**
- **DFS may find a longer or unsafe path.**
- **Greedy Best-First Search may not find the minimum-cost solution.**

Therefore, Uniform Cost Search with Online Search provides the most reliable and cost-effective solution for rescue operations in a partially observable disaster environment.